

Change we can believe in: Comparing longitudinal network models on consistency, interpretability and predictive power



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ABSTRACT

While several models for analysing longitudinal network data have been proposed, their main differences, especially regarding the treatment of time, have not been discussed extensively in the literature. However, differences in treatment of time strongly impact the conclusions that can be drawn from data. In this article we compare auto-regressive network models using the example of TERGMs – a temporal extensions of ERGMs – and process-based models using SAOMs as an example. We conclude that the TERGM has, in contrast to the ERGM, no consistent interpretation on tie-level probabilities, as well as no consistent interpretation on processes of network change. Further, parameters in the TERGM are strongly dependent on the interval length between two time-points. Neither limitation is true for process-based network models such as the SAOM. Finally, both compared models perform poorly in out-of-sample prediction compared to trivial predictive models.

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1. Introduction

The study of social networks is increasingly concerned with modelling network change over time, as longitudinal analysis is usually better equipped for finding explanations and testing theories about the evolution of networks as well as the impact its structure has on constituent nodes (e.g. Steglich et al., 2010). Network analysis over time commonly uses network panel data: a network structure among the same set of nodes that is observed at two or more time points. By now (this is written in early 2017), several statistical approaches are available to analyse such data sets. The most widely used are the stochastic actor-oriented model (SAOM; Snijders et al., 2010b) and several extensions to the exponential random graph model (ERGM; Lusher et al., 2013). These models and variations may appear almost indistinguishable to scientists interested in applying inferential methods to network panel data. However, they rest on quite different statistical assumptions that strongly affect the kind of inference one can draw from the

estimated model parameters and, thus, the kind of questions that can be answered with each method.

While statistical models can be compared on many dimensions, we mainly focus in this article on differences in how they treat time. In particular, we discuss the difference between discrete-time, auto-regressive models and continuous-time, process-based models. Due to its increased use (e.g. in McFarland et al., 2014) and recent claims about its advantage relative to other models for network panel data (Desmarais and Cranmer 2012; Leifeld and Cranmer 2016), we choose the TERGM (or temporal ERGM) for this comparison case to represent auto-regressive models.¹ The continuous-time model we discuss for comparison is the SAOM. Note that for ERGMs both continuous-time and auto-regressive extensions have been proposed – we focus on the latter group. The purpose is to compare the principles of auto-regressive and continuous-time network models and not the relative merits of either particular model – the two cases can be seen as represen-

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¹ We point out commonalities and differences between the TERGM (as defined in Desmarais and Cranmer 2012; Leifeld and Cranmer 2016) and other longitudinal variants of the ERGM where appropriate.

tations of their respective model classes. This article highlights, by way of illustration, the most important differences in assumptions and their interpretive implications between these approaches and thus facilitates the applied researcher's decision which to use in their own research.

1.1. Dimensions of comparison

When comparing statistical models it is tempting to ask which model is “better”. However, “better” implicates at least two quite different dimensions: explanation and prediction. On the one hand, it has been argued that accurate prediction is a chief criterion of a “good” model (Friedman 1953; Jasso 1988). Intuitively a “good” model should be able to extrapolate accurately into the future, which can be tested for a single dataset by simple out-of-sample prediction. At the same time, the criterion for what should be predicted correctly in a model with dependent data (such as networks) is not trivial, as a network is more than just an series of independent tie observations but also the structures that these ties form (see discussion in Section 5).

On the other hand, it has been argued that the endeavour of social science is not to predict, but to explain and understand the world (Hedström 2005; Elster 2007). Models with absurd assumptions or intractable algorithms can generate fairly accurate predictions, but teach us little about the world. Social mechanisms, by contrast, can help us explain the social world and inform our understanding of our own and others' behaviour, but their concatenation in complex ways means that only in the simplest of systems can we expect this to result in accurate prediction at a micro-level. Indeed, even models with poor predictive power can generate valuable insights (see also Epstein 2008). In this line of reasoning, a good model is characterised by reasonable assumptions, as well as by clear interpretability of parameters in light of social mechanisms.

In this paper, we do not necessarily advocate for one or the other position, but investigate how different model assumptions make them applicable to different questions and thus to different empirical problems. As such, we elaborate what conclusions can be drawn from estimated parameters using the SAOM or the TERGM.

The remainder of the article is organised as follows. We first introduce the two different longitudinal/temporal network models (Section 2), and highlight their main features from a statistical point of view. The first main distinguishing feature of the model that is discussed concerns whether it is actor-oriented or tie-oriented (Section 3). Subsequently, the treatment of time is examined. Focus is on the interpretation of parameters and model consistency with regards to the differences between auto-regressive compared to process-based modelling (Section 4). The different treatment of time and how that influences parameters is shown in an empirical example. Finally, we demonstrate that both models perform poorly in out-of-sample prediction (Section 5) across two datasets, suggesting that we need to be careful as to the purposes of longitudinal network research.

2. The models

A social network needs to be understood as a system of interdependent units. Whether one is interested in the details of network dependencies or just needs to control for them, research on networks requires statistical tools that can adequately deal with this challenge. The model families that most explicitly deal with dependencies for such inferential-statistical analysis of social network data are *exponential random graph models* (ERGMs; Frank and Strauss, 1986; Pattison and Wasserman, 1999; Snijders et al., 2006;

Lusher et al., 2013) and *stochastic actor-oriented models* for network evolution (SAOMs; Snijders 2001, 2005; Snijders et al., 2010b).²

2.1. Exponential random graph models

ERGMs were originally formulated for cross-sectional data, i.e., a single observation of a network. The guiding idea behind the model family is to express the probability of observing a given network as a function of subgraphs in this network (called statistics denoted $z(x)$ on the network x), e.g. the reciprocated dyad, or the transitive triplet. These subgraphs express local dependencies between tie variables (reciprocity and transitive clustering, respectively). At the heart of the ERGM lies a linear predictor that weighs the prevalence of statistics in the network by the parameter vector θ :

$$\sum_k \theta_k z_k(x)$$

What is considered local differs between model specifications, with the general rule that specifications including more complex subgraphs instantiate more dependency (Pattison and Snijders, 2013). Model parameters θ_k can be interpreted as expressing, on the tie-level, the probability of observing a specific tie, given the rest of the graph, or on the network-level, as indicating tendencies of a graph to exhibit certain sub-structures relative to what would be expected from a model not containing this parameter (this is discussed further in Section 4).

Longitudinal variants of the ERGM come in two forms, the continuous-time and the discrete-time version. The first, called longitudinal exponential random graph models (LERGMs; Snijders and Koskinen 2013; Koskinen et al., 2015), is a longitudinal, continuous-time form of the ERGM, in the sense that changes to the network are modelled using the conditional probabilities of the ERGM and the process has the cross-sectional ERGM as its limiting distribution. In its treatment of time, the LERGM is identical to the SAOM, thus, we do not focus on the LERGM in this article – the interested reader can generalise from our discussion.

The most prominent discrete-time variant of the ERGM is the temporal exponential random graph model (TERGM; Robins and Pattison 2001; Hanneke, Fu and Xing 2010; Desmarais and Cranmer 2012).³ The model is based on the idea of panel regression. In a sequence of observations, lagged earlier observations or derived information thereof can be used as predictors for later observations. In other words, some of the statistics $z(x)$ are direct functions of an earlier realisation of the network. In its most basic form, the TERGM is a conditional ERGM with an earlier observation of the network occurring among the predictors. It is this basic TERGM (as presented in Desmarais and Cranmer 2012; Leifeld and Cranmer 2016) that we focus on in this article. While other statistics of a previous network realisation (e.g. past two paths) can be included in the model as predictors (e.g. to model transitivity over time), this does not change the fundamental challenges of parameter interpretability or time dependence of parameters modelling dependence as discussed in Section 4; consequently, we only deal with these extended specifications, when necessary, in footnotes. The interested reader can generalise.⁴

² There is a host of models that allow for dependent network ties (such as the p2 model, van Duijn et al., 2004; and an ever-expanding class of latent variable models, see for example the review by Salter-Townshend et al., 2012) that we do not discuss here.

³ Given the limited space in one article, we do not discuss other discrete-time models, such as the StERGM (Krivitsky and Handcock 2014), even though they deserve a similar comparison elsewhere that might give different results.

⁴ It should be noted that the TERGM might only include transformations of an earlier network as predictors of the network, as presented in Hanneke et al. (2010). In this case, all dependence between ties is assumed to be captured by the previous

A different formulation of the basic TERGM does not use a previous time-point as a predictor, but instead models a “dyadic stability” parameter (Leifeld and Cranmer 2016), which models how many ties and non-ties remain constant between two observations. However, as we show in Appendix A.1.1 of the Supplementary data these two formulations are mathematically equivalent and we use the more intuitive version using the previous time-point as a predictor throughout the article for our discussion. As it is just an ERGM with a previous time-point as a dyadic covariate, the TERGM can be estimated with any software that can estimate ERGMs (see e.g. Hunter et al., 2008b; Wang et al., 2014).

2.2. Stochastic actor-oriented models

The guiding idea behind SAOMs is the integration of statistical models that can account for network dependence with theoretical models of action that view social change as emanating from individual actors. Starting from the idea of modelling change, SAOMs are continuous-time models that connect two observations of a network through an unobserved sequence of smallest possible changes, called *mini-steps*. In these *mini-steps*, first an actor in the network is chosen to make a tie change according to the so-called *rate-function*.⁵ Second, this actor considers which (if any) of its outgoing ties it will change, with its decision being based on a multinomial logit that uses the so-called *objective function*. Similar to the ERGM, the *objective function* is a linear predictor that depends on statistics $s_i(x)$ as “seen” from the perspective of actor i and a statistical parameter β :

$$\sum_k \beta_{kS_{i,k}}(x)$$

Network dependencies are modelled within this *objective function* expressed as statistics and, thus, unfold over time, because the decision of which tie to update can depend on its embedding in network structures (such as reciprocity or transitive embedding). Model parameters can be interpreted in light of these *mini-steps*, indicating whether actors preferentially form ties to be embedded in certain configurations. Large values of the objective functions are associated with changes that are more likely.

SAOMs can be fitted to data by means of maximum-likelihood (Snijders et al., 2010a), Bayesian (Koskinen and Snijders 2007), or method-of-moments estimation (Snijders 2001), each implemented in the RSiena software (Ripley et al., 2016). Details of the differences between estimation procedures are immaterial for the arguments presented here unless pointed out explicitly though properties of predictions and efficiency differ between estimation techniques.

3. Actor vs. tie based modelling

The principal difference between any model from the ERGM family and the SAOM family discussed in the literature is that the former is “tie-oriented” while the latter is “actor-oriented” (Block et al., 2016). In a very general sense, this means that the locus of modelling differs between the models. The former models whether

a tie is likely to exist depending on how it is embedded in substructures in the network. This is reflected by the common interpretation of ERGM parameters, which provide the likelihood to observe a tie, conditional on the rest of the network. The latter models whether an actor is more likely to form or maintain a tie depending on its embedding in substructures in the network from the actor’s perspective. The SAOM is explicitly defined on the micro-level to allow for modelling change from an actor’s perspective.

While these differences might seem trivial at first, there are two important implications. First, the different loci of modelling entail different dependence assumptions. In ERGMs, each tie is evaluated on its own for how it is embedded in substructures, but in SAOMs these substructures and ties are always identified and chosen from the perspective of a particular actor. This implies that a decision to, say, create one tie is simultaneously a decision against creating or deleting another at that time, inducing generally higher-level dependence between ties in the SAOM (which might or might not be an appropriate assumption about real-world-processes). Naturally, whether you interpret this ‘decision’ in behavioural terms or merely as formal way of expressing that ties are evaluated with reference to how they are embedded in one actor’s local neighbourhood, is up to the researcher. Second, taking an actor’s perspective allows model specifications that are closer to social theory, as ties in different positions in the same structure can be guided by different model parameters. However, this usually comes at the cost of model parsimony.

Block et al. (2016) discuss the above mentioned differences in detail with extensive illustrations and provide guidelines as to which model might be more appropriate for which research questions when only taking tie- and actor-orientation into account.

4. Process-based vs. auto-regressive modelling

The more important division among the discussed longitudinal network models for the comparison at hand is how they treat time. As argued in Section 2, continuous-time and discrete-time models differ fundamentally: the former models a process whereas the latter models a cross-sectional observation using a previous time-point as a predictor. The conceptualisation of time upon which a model is based strongly impacts how parameters can be interpreted and, accordingly, which kind of research question can be answered.

The difference between process-based and auto-regressive models focussing on networks is presented in Section 4.1. Subsequently, we discuss any potential micro-level interpretations of parameters in connection to underlying social mechanisms (Section 4.2) and the dependence of parameter sizes on time elapsed between two observations (Section 4.3).

4.1. Models of change and models of structure

Continuous- and discrete-time models have different objects of inference and thus answer different types of questions. Continuous-time models like the SAOM answer, broadly speaking, questions about change such as: “According to which regularities does the network evolve from time t_{m-1} to t_m ?”. Conversely, discrete-time models like the basic TERGM answer questions about structure such as: “What regularities does the network at time t_m exhibit, taking into account knowledge about t_{m-1} ?”. Note that this means that the TERGM, despite including a previous realisation of the network, cannot make inference about change. Fig. 1 provides an example that illustrates why it cannot.

Consider modelling the two depicted networks $x_a(t_1)$ and $x_b(t_1)$ using a TERGM with three parameters. We include $x(t_0)$ as a dyadic covariate, a density parameter and a transitive triplets parameter.

time-point; thus, ties in the network under analysis are assumed to be independent. This allows estimation as a simple logistic regression. However, a previous article (Lerner et al., 2013) has already treated this specific model in comparison to the process-based SAOM, concluding that this simplifying assumption leads to reasonable model fit in case observations are temporally very close, i.e. do not differ on a lot of tie variables. Consequently, in this article we focus on model specifications in which previous observations, as well as contemporaneous dependence terms are used to model the network.

⁵ In most applications of SAOMs the rate-function is constant, meaning that an actor is chosen at random.

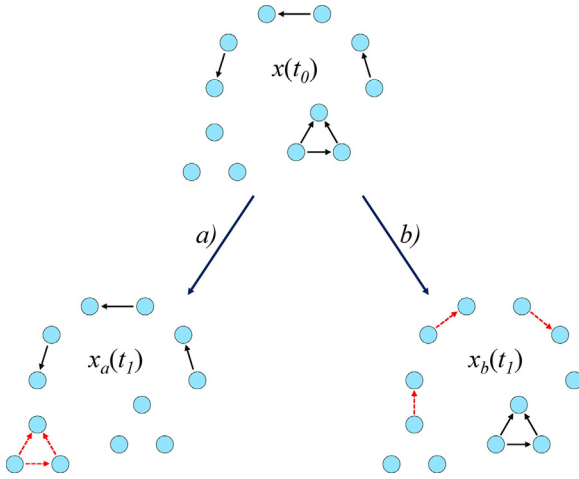


Fig. 1. Two potential network sequences.

Notes: Example of two network realisations $x_a(t_1)$ and $x_b(t_1)$ that are equally likely under the TERGM but imply different change processes for the SAOM given $x(t_0)$. Either network contains 6 ties, 1 transitive triplet and 3 ties change. Solid black arrows indicate stable ties, dotted red arrows indicate changing ties.

Recall that the probability to observe either network $x_a(t_1)$ or $x_b(t_1)$ only depends on the sufficient statistics of the networks:

$$p_{\theta}(x_*(t_1)|x(t_0)) = \frac{\exp(\sum_k \theta_k z_k(x_*(t_1)))}{\kappa}$$

As both, $x_a(t_1)$ and $x_b(t_1)$, have identical statistics (6 ties, 3 ties stable from $x(t_0)$ and 1 transitive triplet), they have the same probability to be observed for any combination of parameters and, consequently, the two hypothetical models have the same Maximum Likelihood Estimate. As the transitive triplets parameter in either model will be identical, we can see that it does not relate to *changes* between two time-points at all, but rather to a higher prevalence of a specific *structure* (the transitive triplet), given the other model parameters. Therefore, the notion that the TERGM models change because a previous time-point is included in the model is wrong.

For the SAOM though, estimating the evolution from $x(t_0)$ to $x_a(t_1)$ will give different results than estimating from $x(t_0)$ to $x_b(t_1)$, as the most likely chain of mini-steps that connects $x(t_0)$ to the respective network at the second time-point differs markedly. In one chain the transitive triplet is stable and the three non-embedded ties are broken and three ties are established in a different location, while in the other chain the transitive triplet must be broken and a new transitive triplet established in a different location. Consequently, the parameter estimates for transitive triplets differ considerably, with the transitive triplets parameter being larger in the case when the triplet emerges from network change.⁶

This illustration shows the fundamental difference between these models. In the SAOM dependence between ties unfolds over time – the embedding of ties guides the *changes* that actors make. In the TERGM, modelling of time is decoupled from the modelling of dependence; one parameter models similarity between a network and the same network at a previous time-point (the auto-regressive stability term), with other parameters modelling the prevalence of specific *structures* in this network without taking the past into account. These differences translate into how the model parameters can be interpreted.

4.2. Parameter interpretation

4.2.1. Interpretation of the SAOM

The natural interpretation of parameters in the SAOM (and similarly of other process-based models like the LERGM) is on the micro-level and follows directly from the formulation of the model. Even though the passing of time is explicitly modelled in the SAOM in the rate-function, generally it is the changes individuals make to their network themselves, expressed in the parameters β_k , that are of interest to researchers.⁷

A parameter's direction indicates whether actors in a (hypothetical) mini-step make choices that increase or decrease the statistic associated with the parameter. Referring to the underlying multinomial choice function, parameter sizes can be translated to (conditional) odds ratios. For example, when an actor has the opportunity to make a change and networks $x^{\pm ij}$ and $x^{\pm ik}$ are two possible outcomes referring to changing either the tie to j or to k , then the relative probability to choose $x^{\pm ij}$ over $x^{\pm ik}$ can be calculated by

$$\frac{p(x \rightsquigarrow x^{\pm ij})}{p(x \rightsquigarrow x^{\pm ik})} = \frac{\exp(\sum_k \beta_k s_{i,k}(x^{\pm ij}))}{\exp(\sum_k \beta_k s_{i,k}(x^{\pm ik}))}$$

In case $x^{\pm ij}$ and $x^{\pm ik}$ are identical on all dimensions but, for example, choosing $x^{\pm ij}$ would result in a reciprocated tie whereas $x^{\pm ik}$ would not, all terms but the difference in reciprocity would cancel out and

$$\frac{p(x \rightsquigarrow x^{\pm ij})}{p(x \rightsquigarrow x^{\pm ik})} = \exp(\beta_{\text{reciprocity}})$$

Because the interpretation of SAOM parameters is on the level of the mini-step, it is important that the modelled real-world process is reasonably approximated by a series of mini-steps. Under this assumption, the interpretation of SAOM parameters allows direct inference on whether particular social mechanisms, once translated into a network statistic, underlie the evolution of a network between multiple time-points.

4.2.2. Interpretation of the ERGM

To interpret the TERGM, it is instructive to review how parameters of the regular (cross-sectional) ERGM can be interpreted. For the ERGM it is common to interpret the model in terms of a micro-process (Lusher et al., 2013, Ch 3). That is, one can interpret the structural features of the network as resulting from biases towards particular types of tie-configurations in a hypothetical network formation process. For example, if there is a bias towards reciprocation, then whenever a tie-variable is being 're-evaluated', it is more likely to remain or be created if it is or would be reciprocated. The underlying model casts the network as the result of a process of local evaluations or updates of ties, where a randomly chosen tie variable is considered for update and set to be present with probability

$$p(x \rightsquigarrow x^{+ij}) = \frac{\exp(\sum_k \theta_k z_k(x^{+ij}))}{\exp(\sum_k \theta_k z_k(x^{+ij})) + \exp(\sum_k \theta_k z_k(x^{-ij}))},$$

where $x^{+ij}(x^{-ij})$ is the network x where the tie from i to j is present (absent).

To accommodate this interpretation we must assume that the process is homogeneous and has continued for a sufficiently long

⁶ Note that Maximum Likelihood estimation is necessary for this example calculation.

⁷ Early continuous-time models for networks that are not actor-oriented and focus explicitly on the rate at which a tie changes are the independent arcs, the reciprocity models (Wasserman, 1980; Leenders, 1995) and the pioneering work of Holland and Leinhardt (1977).

time such that the initial state is irrelevant. In other words, to make this micro-level interpretation for a regular ERGM, one is forced to assume that the observed network is in an *equilibrium* state.

If a researcher is not willing to make this equilibrium assumption, the regular ERGM can still be interpreted in terms of over- or under-representation of specific substructures within a network, compared to what would be expected by chance. For example, a positive reciprocity parameter can be interpreted as a network having more reciprocated dyads compared to what would be expected at random, *controlling* for all other included statistics in the model. This second interpretation is equivalent to viewing ERGM as a log-linear model for dependent binary variables (Frank and Strauss, 1986). The dual interpretation option on the micro-level or on the network-level of the ERGM is a useful model feature because it extends the scenarios when ERGM interpretation is plausible. Is this the same for the TERGM?

4.2.3. Interpretation of the TERGM

The network-level interpretation of the basic TERGM is a model for a network at time t_m where, in addition to the dependencies that a regular ERGM can incorporate, one can also test whether ties that were present at t_{m-1} are more likely to be present at t_m . In this way, the interpretation does not differ from a model where we predict one network from another, e.g. predicting treaties between countries from their geographic proximity. In the TERGM, however, the additional network is the same set of relationships, only at a previous time point. Thus, the interpretation of, for example, a positive transitivity parameter in a TERGM would be that there is more transitivity than randomly expected, controlling for how the network looked in the past.

However, by controlling for a past version of the network, the relative importance of transitivity needs to be interpreted *net of* whichever mechanisms created and changed the network to its previous state. Following from the discussion in Section 4.1, the model for the network at $x(t_m)$ is agnostic about every aspect of the structure of $x(t_{m-1})$ other than the presence of individual tie-variables. Thus, if $x(t_{m-1})$ (the past) contains many of the same dependencies as $x(t_m)$ (the present), the size of the parameters modelling these dependencies will reflect this. For example, an interpretation of a positive transitivity parameter might be that there is a tendency towards transitivity *net of* the past network *and* *net of* the tendency of ties to be transitive in the past. It is important to keep in mind that the TERGM's network-level interpretation of parameters does not pertain to change between two time-points, but to the structure of the second time point, controlling for the first.

Further, and worth *special emphasis*, is that a micro-level interpretation of the TERGM similar to the outlined interpretation of the regular ERGM is generally not possible. This is because the micro-level interpretation of the ERGM requires us to make the assumption that the network generating process is in equilibrium or, in other words, independent of its initial (past) state. A contradiction in the TERGM becomes clear here: To give the TERGM a process-interpretation would require that the network $x(t_m)$ is simultaneously in equilibrium (independent of the past) *and* dependent on the past network $x(t_{m-1})$. This logic is obviously inconsistent and thus a micro-level interpretation of the TERGM is impossible.

This shows that if we are interested in understanding how networks evolve and how ties come about on a micro-level, the TERGM is not a useful model, as parameters estimated from it only allow interpretation on the network level and cannot be interpreted as governing a process over time.

4.3. Dependence of parameters on time

To illustrate a further difference between auto-regressive and process-based models, we can draw on past statistical literature (summarised in e.g. Voelkle et al., 2012). This discusses that not only the size of the “stability-parameter” of auto-regressive models, but also most other parameters of these models depend on the time elapsed between two observations of the same system, in our case a network. This is closely related to the issue of interpretability discussed in the previous section: that the parameters of a structural network need to be interpreted *net of* network features present at a previous time-point. In the remainder of this section, time-dependence is discussed referring first to parameters connected to time and, subsequently, to all other parameters.

Both the SAOM and the TERGM include parameters directly connected to how much time (meaning how many tie-changes) has passed between two observations. In the TERGM this is the “tie-stability” parameter or a transformation thereof; in the SAOM this is the rate-parameter.⁸ A larger SAOM rate-parameter means more time has passed, while a smaller TERGM stability-parameter means the second network looks less like the first network, which allows the interpretation that more time has passed. How about other model parameters?

For the SAOM, more time/more change between observations results in more (simulated) instances in which an actor makes a tie-change and these changes are what is modelled. Thus, for example, a positive reciprocation parameter means that newly formed ties are more reciprocated than expected by chance and/or existing reciprocated ties are less likely to be broken. This means for more change between observations, we base the model parameters on more (hypothetical) decisions. Independence of parameter sizes on time elapsed is a consequence of modelling actors' choices conditionally independent of the rate of change in the SAOM.

In contrast, TERGM parameter estimates are *not* independent of the amount of change between observations. Assume, for example, that we have an observed network $x(t_0)$ that exhibits a strong tendency towards reciprocation of ties, and some process by which ties change but the density and level of reciprocation remain constant. If this process starts from network $x(t_0)$ we can, after some time has passed, observe later realisations of the network $x(t_1), x(t_2)$ etc.

If between $x(t_1)$ and $x(t_0)$ little change has happened, in an analysis of $x(t_1)|x(t_0)$ we will not only observe a strong stability parameter, but also a rather weak reciprocity parameter, as including $x(t_0)$ as a predictor, which is strongly reciprocated, will already explain a lot of the reciprocity in the network under analysis $x(t_1)$. However, the further we move away from $x(t_0)$ in time (read: changes in the network) the less reciprocation in network $x(t_s)$ will be explained by the persistence of reciprocity induced by the network $x(t_0)$ and the larger an estimated reciprocity parameter will be. This is because the reciprocity parameter will model the amount of reciprocation *not captured* by the stability parameter. As the amount of reciprocation not captured by tie-stability depends on the time elapsed, the reciprocity parameter itself will also depend on the time elapsed, where a longer inter-observation time will result in a stronger reciprocity parameter. As the same logic can be applied to all parameters in a TERGM, each one potentially depends on the time elapsed between two observations, *even if* the underlying process is time-homogeneous. The longer the time between the two observations, the less the second network will be explained by the first one and the more structural parameters will seem to matter.

⁸ Assuming that we have a model in which the rate of change is constant between all actors.

Table 1
Parameters used for forwards Simulation of the SAOM and TERGM.

SAOM		TERGM	
Effect	param.	param.	Effect
Outdegree	−2.43	−5.38	Edges
Reciprocity	1.35	1.41	Reciprocity
GWESP transitive	0.91	0.40	GWESP transitive
GWESP cyclic	0.34	0.04	GWESP cyclic
Indegree Popularity (sqrt)	0.02	−0.38	Indegree GW
Two-Paths (OutPop) (sqrt)	−0.45	−0.09	Two-paths GW
Outdegree Activity (sqrt)	−0.09	0.10	Outdegree GW
Same Sex	1.18	1.23	Same Sex
Rate	2.63	4.65	Tie-Stability

The dependence of TERGM parameters on time also points to a more general observation: networks can only follow an assumed TERGM at specific points in time that are determined by a constant distance between observations. If the network at t_1 follows a TERGM conditional on t_0 , then a network at some later point e.g. t_2 does not follow any TERGM conditional on t_0 . This is because the network at t_1 enters the normalising constant, which means one cannot easily rewrite $p(x(t_2)|x(t_0))$ as a TERGM. Consequently one cannot, strictly speaking, estimate a TERGM for observations at t_{1+s} and t_0 (this is shown in detail in Appendix A.1.2 of the Supplementary data), as t_{1+s} does not follow a TERGM distribution based on t_0 .

4.4. Empirical illustration

The dependence of parameter size on time for the two models can be demonstrated using simulation-based analyses. One analysis shows the parameter consistency of the SAOM; the other analysis shows that parameters of a TERGM model are affected by the duration of the assumed process. Detailed description of the performed experiment can be found in Appendix A.2.1 of the Supplementary data, in the current section only the intuition of the analyses and the results are presented.

The analysis proceeds in three steps. First, we estimate models based on empirical data to obtain realistic model parameters. The chosen data is a longitudinal friendship network collected in a school cohort in Glasgow in the 1990s from which we use two waves. It is well known and has been used previously in statistical analyses of network panel data (e.g. Steglich et al., 2010). Model specifications are based on recent literature and include terms that are equally available in RSiena and statnet, the software used for this comparison. The estimation of the same data is independently performed using a SAOM and a TERGM. For each of them a set of parameters is obtained. The estimated parameters are shown in Table 1 for either model. While the model specifications are not identical (they cannot be, see Block et al. (2016) and the discussion in Section 5), they include the same number of parameters that express the same types of interdependencies, i.e., fulfil the same functional role. The results are in line with what is known about networks of similar type. As these parameters are only estimated to obtain realistic parameter values for simulation, they will not be discussed further.

Second, those estimated model parameters are used to simulate 100 replicates of 10 waves of data each for SAOMs and TERGMs starting from the respective previous time-point. In both cases it is made sure that the generated sequences are similar in their amount of change to assure comparability. Note that two series of networks are simulated: one using the SAOM to generate a series of data and one using the TERGM to generate a series of data.

Third, we take a pair of two simulated networks that are between one and ten simulated periods apart and re-estimate models with these data. Each model only re-estimates the data that was produced with the same simulation method. Thus, the SAOM estimates data produced with a SAOM, and the TERGM estimates data produced with a TERGM. This is because we are interested in consistency of the models with regards to stability of parameters and want to exclude all other possible factors that can lead to diverging results.

Our theoretical discussions suggest that re-estimated parameters should be consistent with the simulation parameters in case of the SAOM, regardless of how far the observed networks are apart temporally, but that the distance between networks does affect the size of TERGM parameters.

The results of the re-estimation are shown in Fig. 2. For each parameter, the red triangle is the original parameter used to simulate data, while the coloured points are the median re-estimated parameters after 1, 2, ..., 10 periods, respectively. Bars surrounding the points indicate the 90% of the range of parameter estimates in the repetitions. The estimated parameters of the SAOM show some stochastic variation over the different period lengths, but no systematic bias in one or the other direction. In contrast, however, most mean parameter estimates for the TERGM show a systematic and substantial bias in parameter estimates dependent on the number of simulated periods that lie between the two networks under analysis.

Further, after some periods the range of estimated parameters does not include the original data generating parameter anymore for four out of seven parameters, i.e. the data generating parameter is outside of the 90% interval of recovered parameters. However, not all parameters change with period length. This means that in an empirical analysis, depending on the time lag chosen, results will lead to quite different conclusions, in particular when comparing the relative strength of different parameters.

This analysis shows that if we assume that some observed data are a result of a process that works in continuous time, an analysis of these data with a continuous-time model (SAOM) does not need to consider how far apart the network observations are, as parameter estimates are independent of elapsed time. Should we, however, assume the TERGM as a “data-generating process”, recording the network at the “correct” time-points is of crucial importance, as the estimated parameters are strongly biased otherwise. Arguably it is a herculean task to conclusively show that countries, for example, form treaties based on which treaties existed 1 year ago, but not 2 years ago (or 5 years, or 6 months...). However, without strong assumptions about this, parameter interpretation is ambiguous, as analysis of different time-points will lead to different results.

5. Model performance for tie prediction

After outlining issues of time-dependence of parameters that affect interpretation in the previous section, we now turn to discussing predictive power of the two models under analysis. We first discuss prediction of out-of-sample dependence structures more generally with the conclusion that a model specification should be available that gives a reasonable out-of-sample fit for either model. Then we test recent claims that TERGM provides greater predictive power on the tie-level and thus should be preferred for empirical analysis (Leifeld and Cranmer 2016). Our conclusion is that both models are weak predictively and that models for tie dependence are, for principled theoretical and statistical reasons, not suitable for tie-level predictions.

Cross-validation is a powerful technique for assessing the fit of a particular model to data. It is already common practice to evaluate model fit for ERGM (Hunter et al., 2008a; Robins et al., 2005)

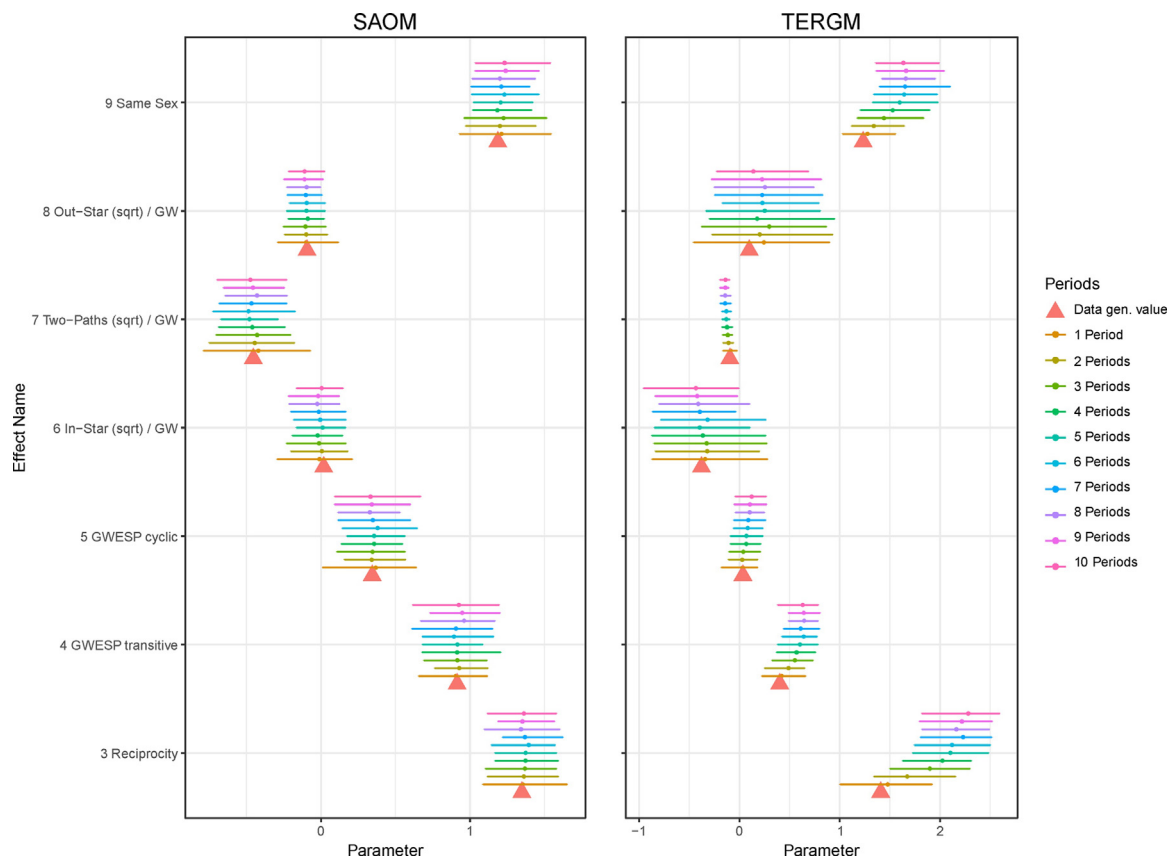


Fig. 2. Estimated Parameters and 90% range of re-estimation of the Glasgow data.

and SAOM (Schweinberger 2012; Lospinoso 2012) through simulating replicate data from the fitted model and comparing those data to the observations. For longitudinal models this can also be done in the form of predictive distributions for out-of sample data (Koskinen and Snijders 2007). For dependent data, especially network data, this begs the question whether a model should perform well in terms of predicting dependence between ties at a future time point, or the presence of individual ties. If tie prediction is perfect, i.e., 100% accurate, it logically implies perfect prediction of all network dependencies. However, this implication breaks down in the absence of perfect tie prediction. What's more, reasonable prediction in terms of tie dependence need not derive from a model's high accuracy at predicting individual ties.

Overall, there seems to be a consensus in the field for the idea that the inferential task of network analysis is to uncover dependencies between network ties; in this view, out-of-sample prediction should be able to predict dependence in un-modelled observations rather than particular observations. The issue is well known more generally in statistics and the parallel to the so called Hamill (2001) forecast – where a forecast is “wrong” but “correct on average” – is particularly relevant. If you calibrate your model ‘marginally’ each prediction of a collection of events is unrealistic but you get the occurrence of each event correct on average; if you calibrate your model dynamically, each predicted outcome is realistic and whether you get the occurrence of each event correctly is not of interest (Gneiting et al., 2007). However, there are also some researchers that advocate for tie-prediction as a model-performance criterion.

When comparing model performance based on *a priori* model specification choices, we must be aware that this choice will likely influence the conclusions on which model performs better. Further, since they are quite different models, it is difficult to translate

model specifications between the SAOM and (T)ERGM for comparing predictive power. Because one is actor-oriented and the other is tie-oriented, the SAOM parameter ‘transitive ties’ does not necessarily model the same as a ‘transitivity’ parameter in an (T)ERGM because the dependence implied differs (see Block et al., 2016). Therefore, model specifications would have to be selected independently, making impartial model comparison difficult due to the practically infinite ways each model can be specified. Thus, we focus here more on the principle considerations of model performance, with special regards to some misconceptions in the literature.

Although assessing predictive power is an inherently practical task, we can derive two expectations on a purely theoretical basis (demonstrated empirically later). First, we can expect both models to perform poorly in tie-level prediction. Second, for the TERGM the inclusion of network terms in the model does not improve the expected performance of the predictive model, unless we take some information about the future into account.

5.1. Theoretical expectations

Both models typically analyse relatively sparse networks of medium size between a few dozen and a few hundred actors. For illustration purposes, let us assume a network of 100 nodes with 500 ties, which results in an average degree of 5. If we assume constant density and a turnover of half the ties (equalling a Jaccard network stability index of 0.33, see Ripley et al., 2016), the predictive task of a model would be to find the 250 ties that will be deleted and another 250 ties that will be created. The probability of an empty model to correctly predict a dropped tie will be 50% (250 out of 500), but the probability to correctly predict a tie that newly comes into existence is only about 2.5% (250 out of 9400). As we know, statistical models in the social sciences have notori-

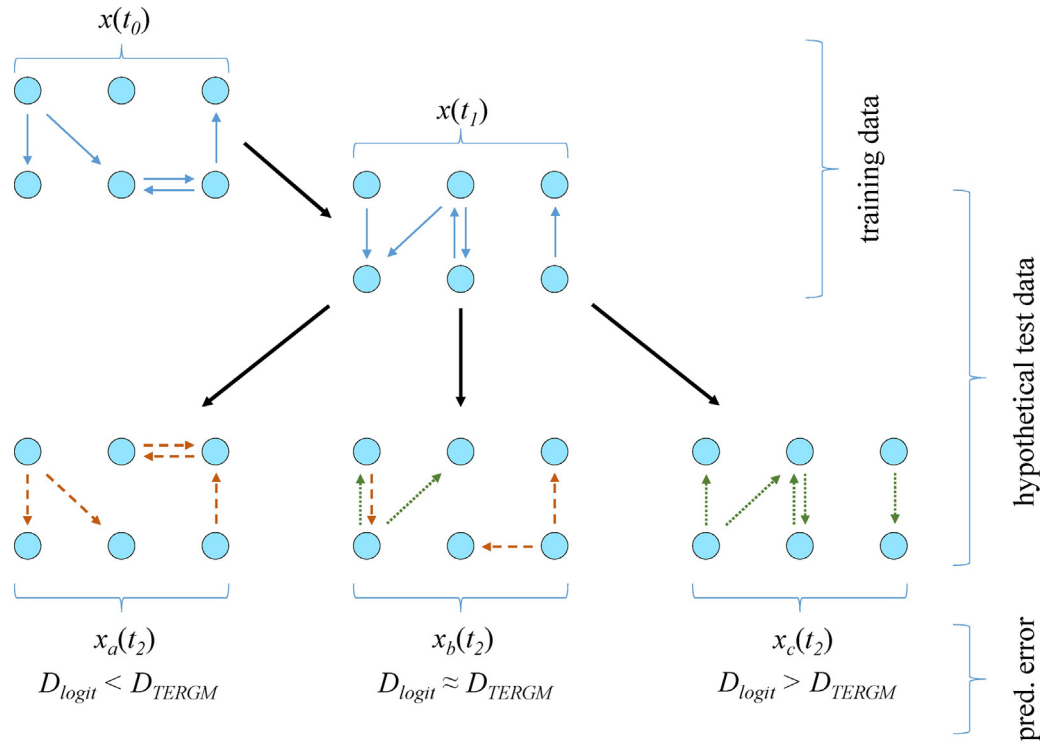


Fig. 3. Potential series of networks at 3 time-points.

Notes: The figure depicts the hypothetical scenario in which a network model is estimated for the evolution from t_0 to t_1 and these parameters are used to predict the network at a future time-point t_2 . Each of the three potential networks $x_*(t_2)$ are equally likely under a TERGM as well as under a logit model, but the tie-level prediction error D differs by network realisation. Brown dashed arrows depict ties that are more likely under the logit model, green dotted arrows depict ties that are more likely under the TERGM.

ously low predictive power, even within sample, as exemplified by typically low R^2 values. Thus, even if we have an unusually good model, we will have difficulties to achieve predictions of tie creations that allow any degree of certainty. Based on probably any model, the answer to the question “will tie XYZ be formed in the future, given that it is absent now?” will always be “probably not (i.e. with a model-based probability of less than 50%)”. Thus, network based predictions about which countries will be the next to form a trade agreement, for example, will usually be bad. We show this in the empirical part by demonstrating that the trivial prediction “the network in the future will look exactly like the network today” greatly and consistently outperforms either model in terms of both, precision (indicating false positives) and recall (indicating false negatives).

The second point mentioned above is more difficult to explain intuitively. It states that a basic TERGM, i.e. modelling tie dependence in addition to tie stability, does not in principle perform better than a logistic regression model that omits all structural network effects in terms of tie-level prediction. This is because the model is agnostic to where the modelled dependence structures are; the model parameter only ensures that a certain number of e.g. reciprocated dyads exist, not that they exist in the right place. Thus, we generally do not improve link prediction unless the inclusion of additional statistics puts further constraints on where these structures are. This can be illustrated on a simple example illustrated in Fig. 3.

Assumes that we estimate a TERGM including a density, a reciprocity and a tie stability term of $x(t_1)$ based on $x(t_0)$ (the training data) and use the estimated parameters to predict any of the networks $x_*(t_2)$ with $x(t_1)$ as a dyadic covariate (the hypothetical test data). It is clear that the process depicted in Fig. 3 is time-homogeneous and that all networks have exactly the same statistics (5 ties, 1 reciprocated tie, 2 ties that existed in the previous wave).

As a measure of tie-level prediction error we use the sum of squared residuals

$$D(p, x(t_2, \cdot)) = \sum_{i,j} (\pi_{ij}^p - x_{ij,t})^2,$$

where p is a probability model (in our case the TERGM) and π_{ij}^p is the probability to observe the tie x_{ij} under this model.

Now an important observation is that $x_a(t_2)$, $x_b(t_2)$ and $x_c(t_2)$ are all equally likely under the TERGM. In other words, we have no reason to believe that either network will be formed more likely than another, given our model. However, the tie-level prediction error between the three networks varies greatly. Compared to a logit model that only includes a density parameter and the previous time-point (i.e. the TERGM from above without structural effects), sometimes the predictive performance of the TERGM is better (e.g. $x_c(t_2)$), sometimes the logit performs better (e.g. $x_a(t_2)$) and sometimes performance is similar (e.g. $x_b(t_2)$). This is because some ties (e.g. from the upper right to the lower right node) are more likely to exist under the model, even though they were not there at the previous time point ($x_c(t_2)$). Thus, if a tie at a later time-point happens to be in this place, the TERGM prediction will be better than a logit model. However, under the TERGM, the network $x_a(t_2)$ is just as likely as $x_c(t_2)$, but here the newly emerged ties are in a location in which there is less predicted weight on this specific location (from the upper middle to the upper right node) – making the prediction of the TERGM worse than the prediction of the logit model. While sometimes one and sometimes the other model performs better, it is usually not possible to know beforehand whether in a particular case the TERGM will improve tie-level prediction. This is because the realisations of either network is equally likely under the model

Table 2
Estimated Parameters used for out-of-sample prediction for the fish data.

	SAOM			TERGM			
	<i>est</i>		<i>s.e.</i>	<i>est.</i>		<i>s.e.</i>	
Rate	1.12	***	(0.28)	8.23	***	(1.10)	Stability
Density	−1.97	**	(0.66)	−15.76	***	(3.30)	Density
GWESP	1.13	**	(0.38)	0.73	***	(0.20)	Alternating Triads
4-Cycles	−0.11		(0.11)	0.01		(0.07)	4-Cycles
Indegree Popularity (sqrt)	0.87	*	(0.31)	−0.27		(0.27)	Alternating In-stars
Contiguity	2.05	***	(0.38)	2.35	***	(0.35)	Contiguity
Same IGOS	0.03		(0.02)	0.03		(0.02)	Same IGOS
Trade (log)	−0.16		(0.10)	−0.20	*	(0.08)	Trade (log)
GDP	2.47	*	(1.19)	2.44	**	(0.77)	GDP
GDP Similarity	−0.90		(1.04)	1.59		(0.95)	GDP Difference
GDP PC	−0.25		(0.20)	−0.04		(0.12)	GDP PC
GDP PC Similarity	−0.63		(0.71)	0.14		(0.18)	GDP PC Difference

that the researcher chose – the TERGM.⁹ This is closely linked to the observations in Section 4.1; the TERGM is only concerned with sufficient statistics, not with the location of where these ties are.

Readers familiar with multi-level modelling will see parallels in the outlined reasoning (see e.g. [Snijders and Bosker 2012](#)). In multilevel models, the inclusion of group level random terms cannot improve prediction of individual outcomes unless other explanatory variables with group-level variance are included in the model. This discussion illustrates that models designed to deal with interdependence between observations are not geared towards predicting (future) outcomes of individual variable values, as this prediction, to make use of the dependence terms, requires knowledge of the (future) outcomes of other variable values. Since we cannot see partially into the future to gain some observations to anchor our inference about what value other, dependent observations may take, we are restricted to inference about the pattern those dependencies have and are likely to take.

5.2. Empirical illustration

In this second empirical illustration of the article, we demonstrate that (these) network models that aim at modelling dependence structures in social networks are not well suited for predicting future outcomes. As above, we only present the intuition of the analyses and the results here; for detailed description of the experiment, refer to Appendix A2.2 of the Supplementary data.

The empirical illustration is straightforward. We take three waves of empirical data from two different subject fields. First, we take one cohort from the ASSIST data ([Steglich et al., 2012](#)), a friendship networks among 80 adolescents, recorded at 3 time-points one year apart, which is representative of data typically analysed in the context of friendship studies. The network density is approximately constant over the three waves. The second dataset is composed of countries' bilateral fisheries treaties as used in [Hollway and Koskinen \(2016\)](#). Ties exist where a treaty concerning the allocation of shared fish stocks or access to fish stocks within the jurisdiction of only one of the parties has been concluded. While time-stamped data is available, we artificially create 3 waves of data in such a way that there is similar amount of change between waves 1 and 2 compared to between waves 2 and 3. The network

evolution mainly consists of new treaties being formed and only few broken, i.e. the density increases over time. This is typical of many political science/international relations datasets.

Using these datasets, we fit a SAOM and a TERGM to the first two waves of the data, with model specifications mirroring specifications in current literature. Results of these estimations can be found in [Tables 2 and 3](#). Based on the estimated models we simulate 1000 networks starting from wave 2 and evaluate the predictive power with regards to wave 3 based on the two criteria *precision* and *recall*. Precision measures the number of correctly predicted ties (simulated ties that are present in the empirical wave 3) over the total number of ties predicted by the simulation. Recall measures the number of correctly predicted ties over the total number of ties observed in wave 3. Perfect prediction would result in both indices being 1; a low value indicates poor prediction. We compare the distribution of precision and recall in 1000 simulated networks for the SAOM and the TERGM to the prediction and recall of a trivial model that assumes that the network in wave 3 will just be the same as the network in wave 2. This is known as the persistence technique in weather forecasting (the weather tomorrow will be like the weather today) – a usually not very accurate and certainly not very useful (yet simple) technique that is often considered as a baseline prediction. A further comparison is the simple logistic regression that excludes all dependence terms, but maintains the information on all covariates in the model.

Results are presented in [Fig. 4](#). The y-axis denotes the value for precision and recall of the 1000 predictions for either model, the different models are strung on the x-axis. It can be seen that neither model predicts better than the simple persistence model, suggesting that neither should be used for tie-level prediction. In fact, both models are, on average only marginally better than the logistic regression that does not use any tie-dependence for prediction and the 95% confidence intervals overlap in almost all cases for the different models. These results are in line with our intuition that predicting rare events, such as the creation of new ties in a sparse network is extremely difficult in a regression framework. In this light, we view any differences in prediction between the TERGM and the SAOM as irrelevant. A further discussion of the results, including the differences between the logit prediction and the TERGM prediction can be found in Appendix A.2.2 of the Supplementary data.

6. Discussion and conclusion

Several approaches to analysing longitudinal network data have been proposed in the literature on statistical network modelling recently. In this article, we compared two of these models that are being applied by practical researchers – the SAOM and the TERGM.

⁹ It should be noted that this limitation does not hold true for process-based models (such as the SAOM), as these consider the embedding of ties in the updating process. This is because the probability to observe a certain network is not a function of network configurations, but of how likely the process that connects two time-points is. Thus, for example, reciprocated ties are more likely to remain than non-reciprocated ties.

Table 3
Estimated Parameters used for out-of-sample prediction for the ASSIST data.

	SAOM			TERGM			
	<i>est.</i>		<i>s.e.</i>	<i>est.</i>		<i>s.e.</i>	
Rate	10.02	***	(1.20)	1.90	***	(0.14)	Stability
Density	−2.20	***	(0.33)	−5.49	***	(0.66)	Density
Reciprocity	2.60	***	(0.28)	2.36	***	(0.30)	Reciprocity
GWESP trans	1.89	***	(0.16)	1.35	***	(0.10)	Alternating Trans. Trip
GWESP trans * Recip.	−1.12	***	(0.33)	−0.42	***	(0.09)	Alternating Cyclic. Trip
Indegree Popularity (sqrt)	−0.06		(0.16)	0.25		(0.32)	Alternating In-stars
Outdeg. Popularity (sqrt)	−0.50	**	(0.17)	1.18	**	(0.35)	Alternating Two-Paths
Sex Alter	0.13		(0.14)	−0.51	*	(0.20)	Sex Alter
Sex Ego	−0.10		(0.16)	−0.81	**	(0.27)	Sex Ego
Same Sex	0.46	**	(0.15)	1.40	***	(0.32)	Sex Interaction

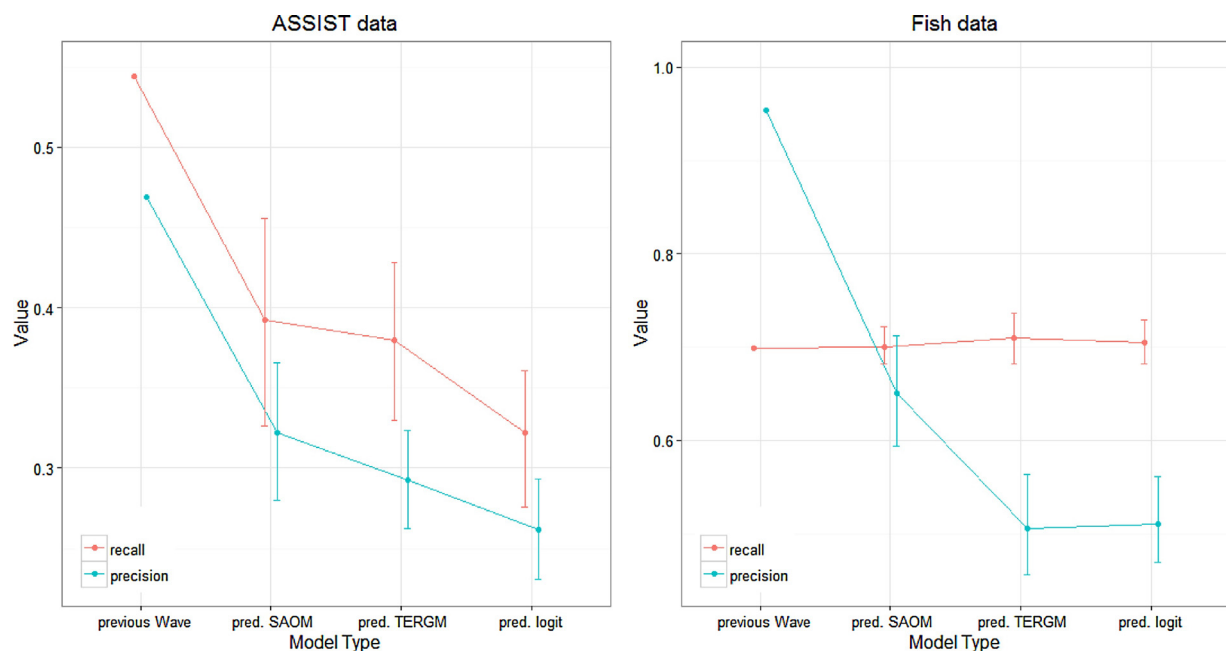


Fig. 4. Predictive performance for out-of-sample tests of SAOM and TERGM for ASSIST data and Fish data.

The former is a process-based, continuous-time model in which dependence unfolds over time, while in the latter, network dependence is modelled within an observation and decoupled from time. This means that the SAOM, and other process-based models, analyse *change* and network *evolution*. The TERGM, and models that treat time and tie dependence similarly, do not analyse change in the included dependence parameters, but can only show that some *structures* exist more than expected by chance, *controlling* for the past where the past carries some of the dependencies. Parameter sizes thus have no consistent micro-level interpretation. This paper shows that including multiple waves of network data in a statistical model is not sufficient to constitute a longitudinal network model allowing inference about change processes.

One implication from this discussion is that, when working with more than two data points, the TERGM (implicitly) requires equidistant observations. Otherwise, the interpretation of lagged predictors across data periods becomes rather incoherent. Deciding whether equidistant observations constitute network data where, between waves, the same amount of time has passed, or whether the same amount of network change has materialised is a further complication.

When considering predictive capabilities, the discussions in this paper suggest that (i) evaluating discrete-time statistical network models that include dependence terms within the second time-

point based on tie-level predictive power is debatable, as they generally do not improve over non-network models.¹⁰ As an illustration of the modest predictive performance we show that (ii) both discussed network models perform poorly when used for tie-level prediction. We think this warrants the conclusion that predictive power is not a useful criterion for discriminating between these types of model. If one believes that prediction is a chief objective of modelling, one might look to a different strand of literature on link-prediction, originating in the physical sciences (see Lü and Zhou, 2011) and not aiming at *explaining* processes of change. Today, experts in machine learning can efficiently solve such prediction tasks, but high predictive accuracy comes at the cost of little insight (Athey, 2017; Breiman, 2001).

Our conclusion that TERGM parameters strongly depend on the length of the interval between two waves have an interesting connection to the findings of Shalizi and Rinaldo (2013). They show that ERGM parameters for two networks of different size have to be different for most cases relevant to empirical researchers. These findings were previously suggested by e.g. Hunter and Handcock

¹⁰ These conclusions apply to the models as currently used in the literature; in case dependence terms that explicitly model the types of ties that are created or maintained are included, the conclusion are likely to differ.

(2006) and equally apply to TERGMs and SAOMs. We add to these findings of non-scalability in terms of system size to by a temporal dimension. In a concise way, our findings show that the SAOM is temporally scalable, while the TERGM is not.

In this article we mainly focus on principle considerations of process-based and auto-regressive models; thus, we neglect some further distinguishing features of these models. While not directly related to the theoretical foundations of the model that we focussed on here, there are further points that will be especially relevant to applied researchers. Future research should address these issues in more detail, for example how easily constraints imposed by the data can be incorporated. Changing composition of a network happens regularly, especially in political sciences, where political actors are newly formed, split or dissolve between observations. For process-oriented models in which the passing of time is considered, changing composition can be taken into account by creating a new actor at some exogenously defined time-point during the modelled process. Discrete-time models currently lack a comparable solution.

This is tightly linked to modelling of multiple dependent variables, for example modelling a network and actor attributes interdependently, or modelling multiple relations between the same or interlocking set(s) of actors. A continuous-time approach can model the reciprocal influences between these dependent variables naturally, as the idea of mini-steps allows an understanding of how change in one variable can lead to change in another, potentially creating feedback-mechanisms. In discrete-time models it is not possible to take changes that occur between observations into account when considering how multiple variables mutually affect one another. Discrete-time models have difficulty instantiating *feedback processes* happening in-between observation moments, thereby missing a lot of what makes network research interesting and worth the effort.

Finally, some notes on processes that directly violate the assumption of stepwise updating of a network are warranted. This is the case, for example, when a group of actors decides jointly and interdependently to form ties amongst themselves or to a third party. Continuous-time models that are based on tie changes one at a time, like the SAOM, have no current implementation that allows this (it should be noted that the framework of the SAOM does not forbid such an extension, though). However, the TERGM has no clear advantage here either. The only dependencies that the TERGM affords are the standard ERGM dependencies and these are not well-equipped to explain ties forming as a result of coordinated action but would, in terms of interpretation, have to be interpreted in terms of how the ties formed conditionally on the others (like the SAOM, the ERGM does not generally forbid such extensions). Thus, while the TERGM does not explicitly disallow coordinated action, it also has no way of accounting for coordinated action.

6.1. Advice for model selection

As outlined in the introduction, model performance in general can be evaluated on the basis of explanatory or predictive power. Recent model comparisons had considered relative out-of-sample predictive power a valid criterion of comparison (Leifeld and Cranmer 2016). However, such comparisons ignore the fact that both models suffer – much like most statistical models in use for studying social phenomena – from their inability to predict rare events such as the creation of new ties in a sparse network. Because a trivial persistence model consistently outperformed both longitudinal network models, we do not believe predictive power on the tie-level a useful criterion for model comparison. This empirical problem is in addition to the theoretical issues with evaluating models based on their predictive capabilities, as outlined in the

introduction.¹¹ This leaves explanatory power as a criterion for model selection.

Here the advice is very clear and follows directly from the discussion in Section 4. Is a researcher interested in explaining the evolution, i.e. change in a network between two time-points, or interested in explaining the structure of an observed network? In the former case we believe we have made a compelling case that process-based models, such as the SAOM (or the LERGM) are preferable, given that they directly model a process, which results in both consistent model parameters independent of the duration of the underlying process and a meaningful micro-level interpretation, allowing direct inference on underlying social mechanisms.

If a researcher is interested in explaining the structure of a network, the cross-sectional class of ERGMs is a well-established starting point. If additionally the previous states of the same network should be taken into account, the TERGM might be used, but its limitations in parameter interpretability and consistency should be considered. When a researcher is interested in whether modelled network dependencies are present in a network beyond dependencies that are solely a relic from the past, one could for example compare results of a TERGM and a cross-sectional ERGM of the second network.

Our critique of the TERGM's parameter interpretability and consistency does not mean that discrete-time network models in general should have no place whatsoever in the researcher's toolbox. In clearly round-based network-evolution processes, for example experimental research in lab-settings, or when knowledge about network ties is made public only at specific, equidistant times, treating this as evolving in discrete time may be warranted. However, for these types of data models like the proposed TERGM-variant by Hanneke et al. (2010), might be used, where whether a tie exists in this round is *only* dependent on structures of the network in previous rounds, thus marrying time and tie dependence in a (maybe) more meaningful way.

6.2. Conclusion

Overall, the discussion in this paper shows that there is a mismatch between some of the research questions that are typically posited for longitudinal networks and the kind of substantive research questions that can be answered using a discrete-time model (such as TERGM). If our goal is to understand change in the world around us, we strongly advocate for statistical models that *model the processes* of change we observe in the world around us. This allows a direct test of theories in the social sciences and inference on how networks unfold over time. This is especially important, given the focus on using social mechanisms to explain observed social phenomena. Whether these processes are best represented by actor-based models (such as the SAOM) or tie-based models (such as the LERGM) is a different matter. In case a researcher assumes a completely different process, not based on sequential, myopic changes, but based on for example co-ordinated or strategic action, we strongly advocate a principled approach that involves thinking about the assumed process and which established model represents this process best. There is, of course, always the possibility to develop a model tailored to specific applications and theoretical assumptions. We are looking forward to future developments that advance our understanding of network change processes of various kinds.

¹¹ However, this position might require further treatment if model specifications and empirical cases are found for which good predictive power of either model is achieved. In this case, comparing these models to further ones, for example different latent space models that makes use of a previous time-point and p2 models, is useful.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.socnet.2017.08.001>.

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